

Home

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Audited By

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Unlock Your Home's Energy Potential

Discover how you can save money, improve comfort, and reduce your environmental impact with our comprehensive Home Energy Assessment. This report provides personalized insights and actionable recommendations to help you make your home more energy-efficient and sustainable.

-Contractor Portal Info: energync.goeverblue.com

-Call center 866-998-8555, Monday–Friday, 8 a.m. to 7 p.m.

-If you need to submit missing usage documents please email them to support@energysavernc.org

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We listened to you!

We want to make sure we're covering all your concerns about your home. If we missed anything in this report, please let us know right away.

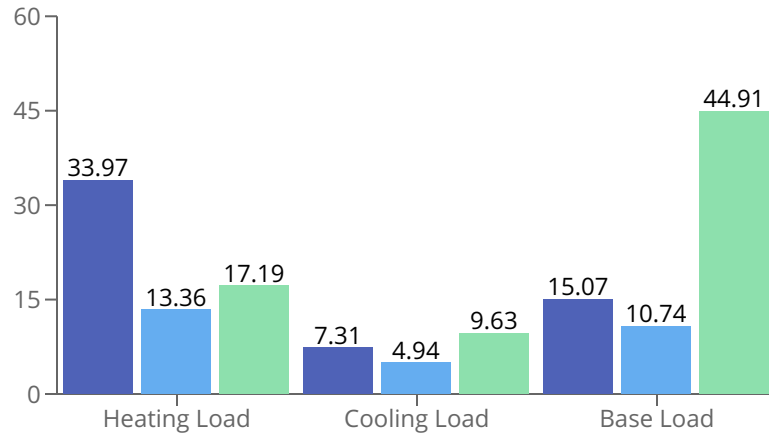
Your Priorities Matter

(Priority 1) What is the main purpose of our visit today

Your Energy History

Yearly Energy Consumption (MMBtu)

● Now ● Goal ● Similar homes



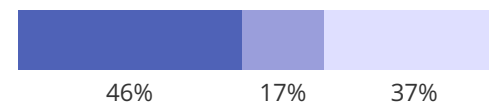
Load Profile (MMBtu)

● Heating ● Cooling ● Base Load

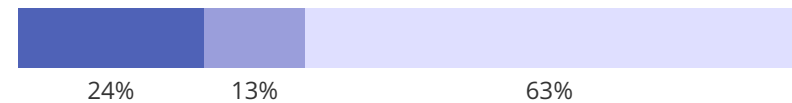
Now: 56.35 MMBtu



Goal: 29.04 MMBtu

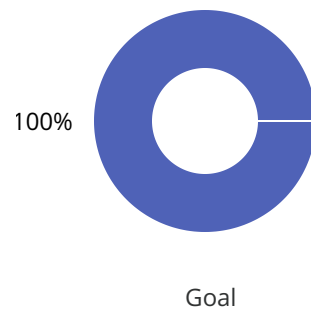
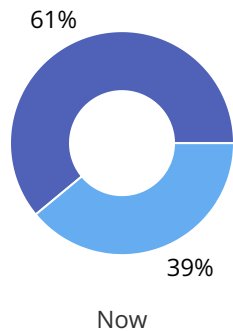


Similar homes: 71.73 MMBtu

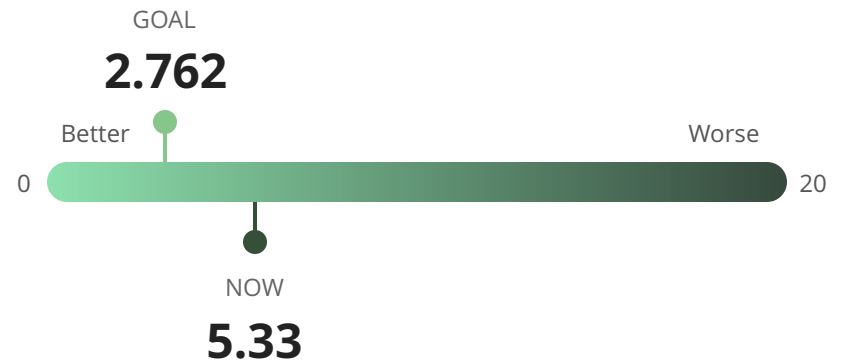


Energy Mix by Fuel Type

● Electricity ● Wood



CO2 Footprint (tons)

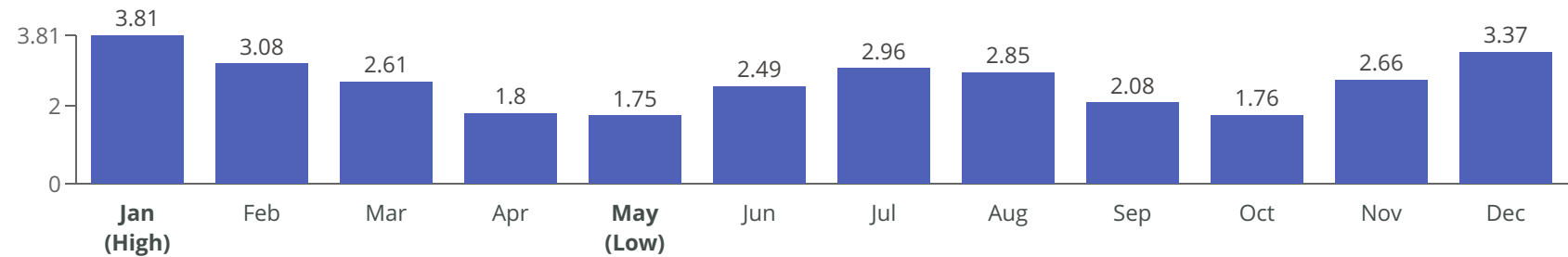


Your Energy History

About these charts

These numbers are weather normalized and then projected based on the 30 year weather normals data from NOAA.
In other words, this is the modeled energy consumption of the home for a typical year, not the year that the utility bills were from.

Electricity Use (MMBtu)



ENERGY SAVER NORTH CAROLINA

TOTALS

Estimated Savings
\$569 per year

This is an estimate of how much you could save starting in Year 1. Savings will only increase as energy prices rise over the years.

Savings to Investment Ratio*
For Package: 1.5

Estimated Incentives
\$ 0

Instant Rebate **\$ 0**

Post Purchase **\$ 0**

Tax Credits **\$ 0**

Impact of upgrades

Electricity Savings **1,592 kWh**

Electricity Reduction **16%**

Carbon (CO2) Savings **3 tons**

Equivalent cars removed from the road **0.5/yr**

Your Home Solutions

Call us today at 866-998-8555 to ask a question or discuss the next step!

DETAILS	APPROXIMATE ANNUAL SAVINGS	SIR *
Upgrade Heating System	\$ 228	100
Upgrade Cooling System	\$ 94	0.6
Seal Air Leaks	\$ 66	1.6
Upgrade Water Heater	\$ 181	1

Total estimated incentives available for this job:



* SIR is the Savings to Investment Ratio. Simply put, if the SIR is 1 or greater, then the energy savings from the item will pay for itself before it needs to be replaced again. This metric is used to help prioritize the recommendations by financial merit.

ENERGY SAVER NORTH CAROLINA

HEATING SYSTEM

Annual Energy Savings

Approx. \$228

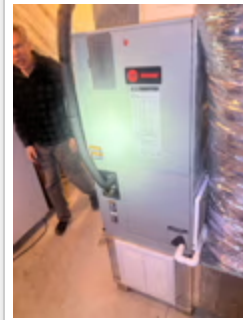
Savings to Investment Ratio

100

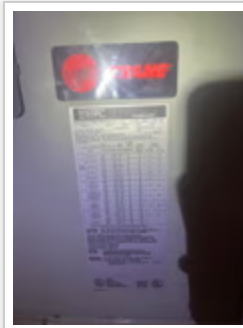
Why it matters

Upgrading to a more efficient furnace, boiler, or heat pump can save you a lot of money, especially if your current one is old. Look for appliances with the ENERGY STAR rating. If you use gas, go for a sealed combustion unit because it's safer—the exhaust goes straight outside.

Upgrade Heating System



Upgrade Heating System - New Heat Pump Model



Consider upgrade on removable non-electric system; Wood Stove and replace on HEAR with mini split

(percentage is still over 20% without added recommendation)

Notes to Homeowners

Upgrading to a high-efficiency ENERGY STAR heating system can save you up to 20% on your heating costs compared to older, less efficient models. Regular maintenance of your current system, such as cleaning or replacing filters and having annual professional check-ups, can improve efficiency by up to 15%. This not only reduces energy bills but also helps prevent costly repairs and extends the lifespan of your heating system. Investing in these practices can lead to significant savings and a more comfortable home.

ENERGY SAVER NORTH CAROLINA

HEATING SYSTEM

Annual Energy Savings

Approx. \$228

Savings to Investment Ratio

100

Why it matters

Upgrading to a more efficient furnace, boiler, or heat pump can save you a lot of money, especially if your current one is old. Look for appliances with the ENERGY STAR rating. If you use gas, go for a sealed combustion unit because it's safer—the exhaust goes straight outside.

...CONTINUED

Upgrade Heating System

DETAILS	NOW	GOAL
Heating System		
Heat Pump System		
System Name	Heat Pump System	
Equipment Type	Central Heat Pump (shared ducts)	
Upgrade action	Replace with a newer model	
% of Total Heating Load	50%	80%
Heat Pump Inverter	No	No
Heating Energy Source	Electricity	Electricity
Heating Capacity	30000 BTU/h	30000 BTU/h
Heating System Efficiency	6.8 HSPF	8.2 HSPF2
Heating System Model Year	1999	2026
Small Wood Stove		
System Name	Small Wood Stove	
Equipment Type	Stove or Insert	
Upgrade action	Remove a system permanently	
% of Total Heating Load	40%	
Heating Energy Source	Wood	
Heating Capacity	24000 BTU/h	
Heating System Efficiency	60 AFUE	
Heating System Model Year	1999	

ENERGY SAVER NORTH CAROLINA

HEATING SYSTEM

Annual Energy Savings

Approx. \$228

Savings to Investment Ratio

100

Why it matters

Upgrading to a more efficient furnace, boiler, or heat pump can save you a lot of money, especially if your current one is old. Look for appliances with the ENERGY STAR rating. If you use gas, go for a sealed combustion unit because it's safer—the exhaust goes straight outside.

...CONTINUED

Upgrade Heating System

DETAILS	NOW	GOAL
Heating System		
Aux Heating		
System Name	Aux Heating	
Equipment Type	Electric Resistance	
Upgrade action	Replace with a newer model	
% of Total Heating Load	10%	5%
Heating Energy Source	Electricity	Electricity
Heating Capacity	30000 BTU/h	30000 BTU/h
Heating System Efficiency	100 AFUE	100 AFUE
Heating System Model Year	1999	2026
Mini Split System (HEAR)		
% of Total Heating Load		15%
Heat Pump Inverter		No
Heating Energy Source		Electricity
Heating Capacity		12000 BTU/h
Heating System Efficiency		8.2 HSPF2
Heating System Model Year		2026

Potential Incentives

INSTANT REBATE

HOMES Income Qualified

HOMES Less Than 80% AMI

\$ 0

Total:

\$ 0

COOLING SYSTEM

Annual Energy Savings

Approx. \$94

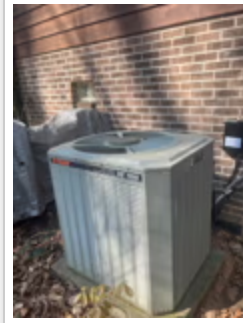
Savings to Investment Ratio

0.6

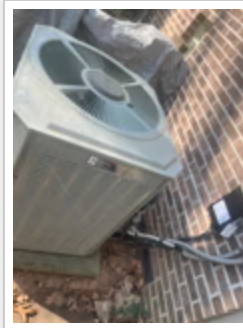
Why it matters

Installing a more efficient air conditioner or evaporative cooler can save you a lot of money, especially if your current unit is old. Look for appliances with the ENERGY STAR rating. If replacing your air conditioner right now isn't practical, plan to get a high-efficiency ENERGY STAR unit (14 SEER or higher) when you can.

Upgrade Cooling System



Upgrade Cooling System - New Heat Pump Model



Notes to Homeowners

Regular maintenance of your cooling system can improve efficiency by up to 15%, which can lead to noticeable savings on your energy bills. For example, cleaning or replacing filters regularly can reduce energy consumption by 5-15%. Annual professional check-ups can help identify and fix issues early, preventing costly repairs and extending the lifespan of your system. These practices not only save money but also ensure your cooling system runs smoothly and keeps your home comfortable.

ENERGY SAVER NORTH CAROLINA

COOLING SYSTEM

Annual Energy Savings

Approx. \$94

Savings to Investment Ratio

0.6

Why it matters

Installing a more efficient air conditioner or evaporative cooler can save you a lot of money, especially if your current unit is old. Look for appliances with the ENERGY STAR rating. If replacing your air conditioner right now isn't practical, plan to get a high-efficiency ENERGY STAR unit (14 SEER or higher) when you can.

...CONTINUED

Upgrade Cooling System

DETAILS	NOW	GOAL
Cooling System		
Heat Pump System		
System Name	Heat Pump System	
Equipment Type	Central Heat Pump (shared ducts)	
Upgrade action	Replace with a newer model	
% of Total Cooling Load	100%	85%
Cooling Capacity	30000 BTU/h	30000 BTU/h
Cooling System Efficiency	11 SEER	15.2 SEER2
Mini Split System (HEAR)		
% of Total Cooling Load		15%
Cooling Capacity		12000 BTU/h
Cooling System Efficiency		15.2 SEER2

Potential Incentives

INSTANT REBATE

HOMES Income Qualified

HOMES Less Than 80% AMI

\$ 0

Total:

\$ 0

AIR LEAKAGE

Annual Energy Savings

Approx. \$66

Savings to Investment Ratio

1.6

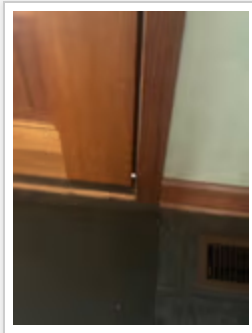
Why it matters

Air sealing is usually the best bang for your buck when it comes to home improvements. To do it right, a big fan called a blower door is used to lower the air pressure in your house. This makes it easy to spot air leaks and fix them. A good air sealing job will make your home much more comfortable and save you a lot of energy.

Seal Air Leaks



Air seal all openings outside conditioned space



Notes to Homeowners

Cutting down on air leaks in your home is a cost-effective way to lower heating and cooling costs. It also improves durability, boosts comfort, and creates a healthier indoor environment.

Now & Goal

DETAILS	NOW	GOAL
Air Leakage		
Blower Door Test Performed	Estimate	Estimate
Blower Door Reading	3490 CFM50	2617.5 CFM50
Conditioned Air Volume	14000 ft ³	
Wind Zone	2	

ENERGY SAVER NORTH CAROLINA

...CONTINUED

Seal Air Leaks

AIR LEAKAGE

Annual Energy Savings

Approx. \$66

Savings to Investment Ratio

1.6

Why it matters

Air sealing is usually the best bang for your buck when it comes to home improvements. To do it right, a big fan called a blower door is used to lower the air pressure in your house. This makes it easy to spot air leaks and fix them. A good air sealing job will make your home much more comfortable and save you a lot of energy.

DETAILS	NOW	GOAL
Air Leakage		
N-Factor	14.99	
Equivalent NACH	1 NACH	0.75 NACH
Effective Leakage Area	192.23 in ²	144.17 in ²
Equivalent ACH50	14.96 ACH50	11.22 ACH50
Mechanical Ventilation Type	None	None
Whole home is professionally air sealed	No	

Potential Incentives

INSTANT REBATE

HOMES Income Qualified

HOMES Less Than 80% AMI

\$ 0

Total:

\$ 0

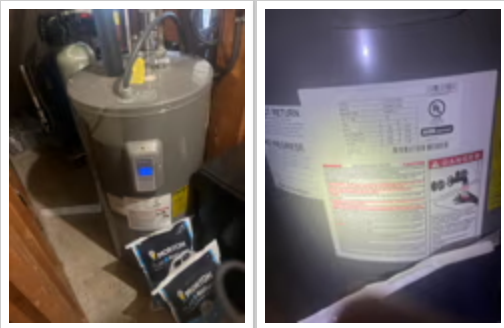
WATER HEATER

Annual Energy Savings
Approx. \$181

Savings to Investment Ratio
1

Why it matters
Switch to a heat pump water heater to save energy and lower the risk of dangerous carbon monoxide leaks in your home if you currently have a gas water heater.

Upgrade Water Heater



Consider an ENERGY STAR certified heat pump/hybrid water heater



Notes to Homeowners

Heat pump water heaters use electricity to move heat from one place to another instead of generating it directly. This makes them two to three times more energy efficient than regular electric water heaters.

Now & Goal

DETAILS	NOW	GOAL
Water Heater		
Water Heater 1		
Fuel	Electricity	Electricity
Type	Tank Water Heater	Heat Pump
ENERGY STAR		Yes

WATER HEATER

Annual Energy Savings
Approx. \$181

Savings to Investment Ratio
1

Why it matters
Switch to a heat pump water heater to save energy and lower the risk of dangerous carbon monoxide leaks in your home if you currently have a gas water heater.

...CONTINUED

Upgrade Water Heater

DETAILS	NOW	GOAL
Water Heater		
Energy Factor	91 EF	333 EF
Model Year		2026

Potential Incentives	INSTANT REBATE
HOMES Income Qualified HOMES Less Than 80% AMI	\$ 0

Total: **\$ 0**

Health & Safety Test

Why it matters

These tests, recommended by the Building Performance Institute (BPI), can help spot potential health and safety issues in your home.

Test Summary

Ambient Carbon Monoxide	✓
Mold & Moisture	⚠
Roof Condition	✓
Drainage System Condition	✓

✓ Passed ✗ Failed ⚠ Warning

Notes to Homeowners

TOTAL SAVINGS

INSTANT REBATES

\$ 0

POST PURCHASE REBATES

\$ 0

TAX CREDITS

PROGRAM WISE TOTAL

HOMES Income Qualified

Potential Incentives

POTENTIAL INCENTIVES	AMOUNT
INSTANT REBATE	\$ 0
HOMES Income Qualified HOMES Less Than 80% AMI	
POST PURCHASE	\$ 0
TAX CREDIT	\$ 0

Total potential incentives:

Terms and conditions apply. All incentives and amounts are shown on a best guess basis. Final amount and availability are subject to change and approval of the incentive program implementer.

ENERGY SAVER NORTH CAROLINA

POTENTIAL INCENTIVES

Attributes for this Incentive

Max amount:

\$ 16,000.00

Max % of cost:

100%

End date:

2029-01-23

Incentive Total:

INCENTIVE • HOMES Income Qualified

HOMES Less Than 80% AMI

Program:

HOMES Income Qualified

Benefit type:

Instant Rebate

Estimated Amount:

Description

DETAILS	COST	INCENTIVE %	INCENTIVE AMT.
Upgrade Heating System	\$ 0	NaN%	\$ 0
Upgrade Heating System	\$ 0	NaN%	\$ 0
Upgrade Cooling System	\$ 2,279.70	NaN%	\$ 0
Upgrade Water Heater	\$ 2,400.00	NaN%	\$ 0
Seal Air Leaks	\$ 636.09	NaN%	\$ 0

Notes

WORK ORDER DETAILS

ID#

390562

Date

__/__/__

Bryan Wussow

6960 Old Ridge Road

Waxhaw, NC 28173

From

ENERGY SAVER NORTH CAROLINA

Work Order

ITEM	QTY	UNIT
Upgrade Cooling System		
Seal Air Leaks		
Upgrade Water Heater		
UPGRADE HEATING SYSTEM		
Upgrade Heating System	1	each

Next Steps After Your Home Energy Assessment

Thanks for completing your home energy assessment. To move forward with the recommended upgrades, you'll need to get in touch with an Energy Saver North Carolina registered contractor. They can help you make the necessary improvements to boost your home's energy efficiency.

CONTACT A CONTRACTOR

If you have any questions or need help finding the right contractor, just give us a call at [866-998-8555](tel:866-998-8555).

ENERGY SAVER NORTH CAROLINA

ABOUT THE METRICS

These metrics are for the whole house in a pre and post-retrofit state.

The 'Base' savings numbers will likely not be the same as the actual energy consumption of the home. These numbers are weather normalized and then projected based on the 30 year weather normals data from NOAA. In other words, this is the modeled energy consumption of the home for a typical year, not the year that the utility bills were from.

Metrics

FUELS	BASE	IMPROVED	SAVED
Total Fuel Energy Usage therms/year	219	0	219
Wood Energy Usage cords/year	1.09	0	1.09

[View detailed usage →](#)

METRIC	BASE	IMPROVED	SAVED
Electric Energy Usage kWh/year	10,101	8,509	1,592
Total Energy Usage MMBtu/year	56.34	29.04	27.30
Fuel Energy Cost \$/year	\$ 328	\$ 0	\$ 328
Electric Energy Cost \$/year	\$ 1,526	\$ 1,286	\$ 240
Total Energy Cost \$/year	\$ 1,854	\$ 1,286	\$ 568
CO2 Production Tons/year	5.3	2.8	2.5
Payback years			8
Total Energy Savings			48.46%
Total Carbon Savings			48%
Net Savings to Investment Ratio SIR			1.5
Net Annualized Return MIRR			8.0%

HEATING & COOLING LOAD CALCULATIONS		
Heating Load Btu/hr	Base: 24,661	Improved: 25,334
Cooling Load: Sensible Btu/hr	Base: 24,132	Improved: 23,597
Cooling Load: Latent Btu/hr	Base: 1,629	Improved: 1,593
Winter Design Temperature	Outdoor: 26°	Indoor: 70°
Summer Design Temperature	Outdoor: 91°	Indoor: 75°

ENERGY SAVER NORTH CAROLINA

Tech Specs

Property Details

Year Built:	1999
Conditioned Area:	1500 ft ²
Area Includes Basement:	Yes
Average Wall Height:	9 ft
Floors Above Grade:	2
Number of Occupants:	2
Number of Bedrooms:	3
Type of Home:	Single Family Detached
Front of Building Orientation:	North West
Shielding:	Normal
Tuck Under Garage:	No

Heating & Cooling

Heating Design Load:	24661 Btu/hr
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Hvac: 1

System Name:	Heat Pump System
Equipment Type:	Central Heat Pump (shared ducts)
Upgrade action:	Replace with a newer model
Heat Pump Inverter:	No
Heating Energy Source:	Electricity
% of Total Heating Load:	50%
Heating Capacity:	30000 BTU/h
Heating System Efficiency:	6.8 HSPF
Heating System Model Year:	1999
Heating System Equivalent Full Load Hours:	585
% of Total Cooling Load:	100%
Cooling Capacity:	30000 BTU/h

Cooling System Efficiency:	11 SEER
Cooling System Equivalent Full Load Hours:	752
Duct	50/50 Attic (unconditioned) - Conditioned
Location:	Space
Duct Insulation:	Reflective bubble wrap
Duct Leakage:	6% - Well sealed
Duct Efficiency:	93.41%

Hvac: 2

System Name:	Small Wood Stove
Equipment Type:	Stove or Insert
Upgrade action:	Remove a system permanently
Heating Energy Source:	Wood
% of Total Heating Load:	40%
Heating Capacity:	24000 BTU/h
Heating System Efficiency:	60 AFUE
Heating System Model Year:	1999
Heating System Equivalent Full Load Hours:	547

Hvac: 3

System Name:	Aux Heating
Equipment Type:	Electric Resistance
Upgrade action:	Replace with a newer model
Heating Energy Source:	Electricity
% of Total Heating Load:	10%
Heating Capacity:	30000 BTU/h
Heating System Efficiency:	100 AFUE
Heating System Model Year:	1999
Heating System Equivalent Full Load Hours:	109

Hvac: 4

System Name:	Mini Split System (HEAR)
Equipment Type:	Ductless Heat Pump
Upgrade action:	Install a new non-existing system

Appliances

Range: 1

Range Fuel Type:	Electricity
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Oven: 1

Oven Fuel Type:	Electricity
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Clothes Dryer: 1

Dryer Fuel Type:	Electricity
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Clothes Washer

Type:	Top Load
Integrated Modified Energy Factor:	0.64 IMEF
ENERGY STAR:	No

Dishwasher

Dishwasher Installed?:	Yes
Energy Factor:	0.55 EF
ENERGY STAR:	Yes

Refrigerators

Refrigerator: 1

Name:	Refrigerator 1
Refrigerator Age:	0-14
Refrigerator Size:	22+
ENERGY STAR:	No
Usage:	826 kWh/yr

Lighting

# of CFLs:	7
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ENERGY SAVER NORTH CAROLINA

Tech Specs

# of LEDs:	14
# of Incandescents:	10
Total # of Light Bulbs:	31

Doors

Door: 1

Type:	1/2-Lite Wood
Area:	21 ft ²
ENERGY STAR:	No
U Value:	0.58 U Value

Door: 2

Type:	1/2-Lite Steel, insulated
Area:	21 ft ²
ENERGY STAR:	No
U Value:	0.26 U Value

Exterior Walls

Wall: 1

Modeled Area:	1717.02 ft ²
Insulated?:	Yes
Siding:	Wood/Fiber Cement siding
Construction:	2x4 Frame
Cavity Insulation:	11 R Value
Continuous Insulation:	0 R Value

Attic & Vaulted Ceiling

Attic: 1

Modeled Area:	500 ft ²
Insulation Depth:	7-9
Insulation	Fiberglass or Rockwool (batts or blown)
Type:	blown
Insulation:	18.9 R Value
Radiant Barrier?:	No

Has Knee Wall?:	No
Cool Roof?:	No

Foundation - General

Foundation: Basement:	100%
Foundation Above Grade Height:	4 ft

Foundation - Basement

Modeled Basement Floor Area:	500 ft ²
Basement Wall Insulation:	Fiberglass blanket
Basement Continuous Wall Insulation:	4 R Value
Basement Rim Joist Treatment:	Same as Basement Wall
Basement Heating:	Intentional
Basement Cooling:	None or Undesired Incidental

Frame Floors

Modeled Floor Area:	0 ft ²
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Windows

Window: 1

Window Area: NE (Left):	29.43 ft ²
Window Area: SE (Back):	147.17 ft ²
Window Area: SW (Right):	55.19 ft ²
Window Area: NW (Front):	122.64 ft ²
Type:	Single pane + storm
Frame:	Wood or metal clad
ENERGY STAR:	No
U-Value:	0.51 U Value
Solar Heat Gain Coefficient:	0.56 SHGC
Window Area: NE (Left) Overhang Depth:	2 ft

Window Area: SE (Back) Overhang Depth:	2 ft
Window Area: SW (Right) Overhang Depth:	2 ft
Window Area: NW (Front) Overhang Depth:	3 ft
Exterior Treatment: NE (Left):	No Treatment
Exterior Treatment: SE (Back):	No Treatment
Exterior Treatment: SW (Right):	No Treatment
Exterior Treatment: NW (Front):	No Treatment

Air Leakage

Blower Door Test Performed:	Estimate
Blower Door Reading:	3490 CFM50
Conditioned Air Volume:	14000 ft ³
Wind Zone:	2
N-Factor:	14.99
Equivalent NACH:	1 NACH
Effective Leakage Area:	192.23 in ²
Equivalent ACH50:	14.96 ACH50
Mechanical Ventilation Type:	None
Whole home is professionally air sealed:	No

Water Heating

Water Heating: 1

Fuel:	Electricity
Type:	Tank Water Heater
Age:	11-15
Location:	Indoors and within heated area
Temperature Settings:	Medium (130-140 F)
Energy Factor:	91 EF

Pool & Hot Tub

Pool:	No
Hot Tub:	No

ENERGY SAVER NORTH CAROLINA

Tech Specs

PV

Pv: 1

Has PV?: No

Misc Electric Loads

Heat Tape Length: 0 ft

Max Amps: 200 amps

Utilities

Utility Price Natural Gas: 1.29 \$/Therm

Utility Price Propane: 3.36 \$/Gallon

Utility Price Fuel Oil: 3.44 \$/Gallon

Utility Price Electricity: 0.1511 \$/kWh

Utility Price Wood: 300 \$/cord

Utility Price Pellets: 0 \$/Ton

Utility Bills

Electric

Electric Utility Provider Name NCSEO

Electric Account Number 001Ui00000sz3xzIAA

1. 06/27/2024 757 kWh

2. 07/30/2024 1153 kWh

3. 08/29/2024 826 kWh

4. 09/27/2024 642 kWh

5. 10/30/2024 851 kWh

6. 11/27/2024 765 kWh

7. 12/30/2024 686 kWh

8. 01/30/2025 725 kWh

9. 02/28/2025 704 kWh

10. 03/29/2025 585 kWh

11. 04/29/2025 543 kWh

12. 05/28/2025 740 kWh

Fuel

Fuel Utility Provider Name

Fuel Account Number

1. --

2. --

3. --

4. --

5. --

6. --

7. --

8. --

9. --

10. --

11. --

12. --

Secondary Fuels

Wood

Start Date 01/01/2024

End Date 01/01/2025

Usage 1 Cords

BPI-2400 Calibration

Calibration status Passed

Electricity

Calibration type: Simple

Heating Bias Error: -10%

Cooling Bias Error: -10%

Baseload Bias Error: -10%

Heating Absolute Error: 334 kWh

Cooling Absolute Error: 202 kWh

Baseload Absolute Error: 416 kWh

Wood

Calibration type: Simple

Heating Bias Error: 3%

Cooling Bias Error: N/A

Baseload Bias Error: N/A

Heating Absolute Error: 8 therms

Cooling Absolute Error: N/A

Baseload Absolute Error: N/A

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About This Report

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NCDEQ HOMES

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Annual Fuel Utilization Efficiency (AFUE) The measure of seasonal or annual efficiency of a residential heating furnace or boiler. It takes into account the cyclic on/off operation and associated energy losses of the heating unit as it responds to changes in the load, which in turn is affected by changes in weather and occupant controls.

Annualized Return The return an investment provides over a period of time, expressed as a time-weighted annual percentage. This is the equivalent annual interest rate you would get if you put the same amount of money spent on the energy upgrade into a savings account.

Asbestos Asbestos is a mineral fiber that has been used commonly in a variety of building construction materials for insulation and as a fire-retardant, but is no longer used in homes. When asbestos-containing materials are damaged or disturbed by repair, remodeling or demolition activities, microscopic fibers become airborne and can be inhaled into the lungs, where they can cause significant health problems.

British Thermal Unit (Btu/MMBtu) The amount of heat required to raise the temperature of one pound of water one degree Fahrenheit; equal to 252 calories. MMBtu = one million Btu.

Carbon Monoxide (CO) A colorless, odorless but poisonous combustible gas with the formula CO. Carbon monoxide is produced in the incomplete combustion of carbon and carbon compounds such as fossil fuels (i.e. coal, petroleum) and their products (e.g. liquefied petroleum gas, gasoline), and biomass.

Cashflow When financing energy efficiency improvements, cashflow is the difference between the average monthly energy savings and the monthly loan payment.

Combustion Appliance Zone (CAZ) A contiguous air volume within a building that contains a combustion appliance such as furnaces, boilers, and water heaters; the zone may include, but is not limited to, a mechanical closet, mechanical room, or the main body of a house, as applicable.

Compact Fluorescent Light bulb (CFL) A smaller version of standard fluorescent lamps which can directly replace standard incandescent lights. These highly efficient lights consist of a gas filled tube, and a magnetic or electronic ballast.

Cubic Feet per Minute (CFM) A measurement of airflow that indicates how many cubic feet of air pass by a stationary point in one minute.

Carbon Dioxide (CO2) A colorless, odorless noncombustible gas that is present in the atmosphere. It is formed by the combustion of carbon and carbon compounds (such as fossil fuels and biomass). It acts as a greenhouse gas which plays a major role in global warming and climate change.

Energy Efficiency Ratio (EER) The measure of the energy efficiency of room air conditioners: cooling capacity in Btu/hr divided by the watts consumed at a specific outdoor temperature.

Energy Factor (EF) The measure of efficiency for a variety of appliances. For water heaters, the energy factor is based on three factors: 1) the recovery efficiency, or how efficiently the heat from the energy source is transferred to the water; 2) stand-by losses, or the percentage of heat lost per hour from the stored water compared to the content of the water; and 3) cycling losses. For dishwashers, the energy factor is the number of cycles per kWh of input power. For clothes washers, the energy factor is the cubic foot capacity per kWh of input power per cycle. For clothes dryers, the energy factor is the number of pounds of clothes dried per kWh of power consumed.

Heating Seasonal Performance Factor (HSPF) The measure of seasonal efficiency of a heat pump operating in the heating mode. It takes into account the variations in temperature that can occur within a season and is the average number of Btu of heat delivered for every watt-hour of electricity used.

Heat Recovery Ventilator (HRV) / Energy Recovery Ventilator (ERV)

A device that captures the heat or energy from the exhaust air from a building and transfers it to the supply/fresh air entering the building to preheat the air and increase overall heating efficiency while providing consistent fresh air.

Light Emitting Diode (LED) Lighting An extremely efficient semiconductor light source. LEDs present many advantages over incandescent light sources including lower energy consumption, longer lifetime, improved physical robustness, and smaller size.

Modified Internal Rate of Return (MIRR) This is your return on investment. Roughly speaking, if you invested the same amount of money for this project (listed on this report as the total cost) into a bank account, your equivalent interest rate from all of the energy savings would be the MIRR.

N-Factor A factor of how susceptible your house is to wind, influenced by weather patterns, location, and the number of floors in the home. Used in the calculation of NACH.

Natural Air Changes per Hour (NACH) The number of times in one hour the entire volume of air inside the building leaks to the outside naturally.

Payback Period The amount of time required before the savings resulting from your system equal the system cost. Our payback calculation is amortized with an annual inflation of 3% and a fuel cost escalation of 5%.

R-Value A measure of the capacity of a material to resist heat transfer. The R-Value is the reciprocal of the conductivity of a material (U-Value). The larger the R-Value of a material, the greater its insulating properties.

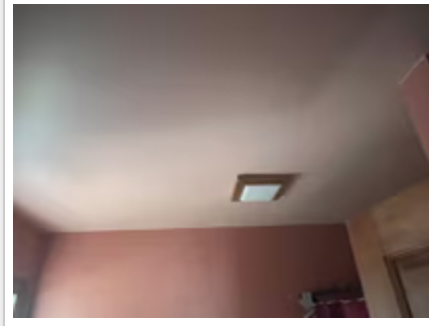
Radon A naturally occurring radioactive gas found in the U.S. in nearly all types of soil, rock, and water. It can migrate into most buildings. Studies have linked high concentrations of radon to lung cancer.

Rim Joist In the framing of a deck or building, a rim joist is the final joist that caps the end of the row of joists that support a floor or ceiling. A rim joist makes up the end of the box that comprises the floor system.

Seasonal Energy Efficiency Ratio (SEER) A measure of seasonal or annual efficiency of a central air conditioner or air conditioning heat pump. It takes into account the variations in temperature that can occur within a season and is the average number of Btu of cooling delivered for every watt-hour of electricity used by the heat pump over a cooling season.

Savings to Investment Ratio (SIR) A ratio used to determine whether a project that aims to save money in the future is worth doing. The ratio compares the investment that is put in now with the amount of savings from the project.

Mechanical Ventilation



HEAR - Upgrade Venting system - Vents into attic

Why it matters

Tracking electric use from things like charging your electric vehicle and using de-icing cables helps you understand where your energy is going. By knowing this, you can find ways to save energy and reduce your electric bill.

Track Electric Usage



HEAR - Pannel and Wiring